Assignment 2: Data Cleaning and Transformation

Dataset Overview & Missing Value Treatment Strategy

ASSIGNMENT QUESTIONS

- **Handling Missing Values (Price):** Check for missing values in the 'Price' column. How would you handle products with missing price information?
- **Handling Missing Values (Category):** If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

1. Complete Dataset (Rows 1–34)

#	Product ID	Product Name	Brand Name	Price	Quantity	Category (F)	Price range Imputed (G)	Category Imputed (H)
1	28-JAN-US	laptop	Dell	1000	30	Electronics	1000	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	130	Kitchen
4	11-APR-US	smartphone	Samsung	900	25	Electronics	900	Electronics
5	22-MAY-US	Backpack	North Face	70	20	*[Blank]*	70	Electronics
6	07-JUN-UK	Headphones	Sony	*[Blank]*	45	Electroni	309	Electroni
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	*[Blank]*	90	Electronics
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	*[Blank]*	120	Electronics
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni	150	Electroni
15	16-APR-ES	headphones	Bose	250	20	Electroni	250	Electroni
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni	160	Electroni
18	27-JAN-IT	laptop	Asus	980	10	Electronics	980	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	150	Fashion
20	14-AUG-US	Camping Tent	Coleman	*[Blank]*	10	Outdoor	309	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics	700	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	80	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	*[Blank]*	150	Electronics
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics	800	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban	*[Blank]*	25	Fashion	309	Fashion
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	400	Kitchen
28	16-APR-ES	headphones	Bose	250	20	Electroni	250	Electroni
29	07-MAR-CA	Dress	Zara	60	30	Fashion	60	Fashion
30	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	40	Kitchen
31	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	130	Electronics
32	02-DEC-CA	Jeans	Levi's	50	50	Fashion	50	Fashion
33	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
34	09-JUL-FR	Watch	Casio	100	20	Accessories	100	Accessories

2. Handling Missing Values Analysis & Strategies

A. Question 1: Missing Values in 'Price'

Audit: 3 products have missing prices (Sony Headphones, Coleman Camping Tent, Ray-Ban Sunglasses).

Spreadsheet Formula Used: `=ROUND(IF(ISBLANK([@price]),AVERAGE([price]),[@price]),0)`

Evaluation & Recommendations:

- **Global Average ($309):** Replaces blanks with overall dataset mean, which distorts specific item values (e.g., pricing a tent equal to luxury sunglasses).
- **Category-Based Mean:** Calculate average price within each product category (e.g., mean price of *Fashion* for sunglasses).
- **Historical / Match Lookup:** Match against identical product entries (e.g. Bose headphones at $250).

B. Question 2: Missing Values in 'Category'

Audit: 4 products have missing category values (North Face Backpack, Adidas Sneakers, Nespresso Coffee Maker, Xiaomi Fitness Tracker).

Spreadsheet Formula Used: `=IF(ISBLANK([@Category]),"Electronics",[@Category])`

Evaluation & Recommendations:

Strategy	Description	Example Outcome
Static Fallback (Current)	Fills missing entries with "Electronics"	Mistakenly marks Backpack & Coffee Maker as Electronics
Keyword Mapping	Derive category from `Product Name`	"Coffee Maker" → Kitchen, "Sneakers" → Fashion
Brand Lookup	Map brand name to main product line	North Face → Outdoor, Nespresso → Kitchen
Text Clean-up	Fix spelling inconsistencies	Normalize "Electroni" → "Electronics"

Assignment 2: Data Cleaning and Transformation

Dataset Report & Data Audit Log

Dataset Record Summary: Displaying 34 extracted product records including Product ID, Brand, Price, Quantity, and Category classifications.

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
1	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	130	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics	900	Electronics
5	22-MAY-US	Backpack	North Face	70	20		70	Electronics
6	07-JUN-UK	Headphones	Sony		45	Electronics	309	Electronics
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40		90	Electronics
13	08-DEC-DE	Coffee Maker	Nespresso	120	35		120	Electronics
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	150	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessorie
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	160	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronics	980	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	150	Fashion
20	14-AUG-US	Camping Tent	Coleman		10	Outdoor	309	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics	700	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	80	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30		150	Electronics
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessorie
25	29-SEP-CA	Smartphone	Google	800	45	Electronics	800	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	309	Fashion

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	400	Kitchen
28	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
29	07-MAR-CA	Dress	Zara	60	30	Fashion	60	Fashion
30	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	40	Kitchen
31	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	130	Electronics
32	02-DEC-CA	Jeans	Levi's	50	50	Fashion	50	Fashion
33	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
34	09-JUL-FR	Watch	Casio	100	20	Accessories	100	Accessorie

Assignment 2: Data Cleaning and Transformation

Dataset Report & Transformation Audit Log

Transformation Log Summary: Captured 3 sequential Find & Replace cleaning operations performed across the dataset to standardize Product Names and Category values.

1. FIND & REPLACE OPERATIONS LOG

STEP	TARGET COLUMN	FIND WHAT	REPLACE WITH	PURPOSE / RESULT
1	Product Name	`headphones`	`Headphones`	Standardized lowercase product entries to Title Case
2	Product Name	`smartphone`	`Smartphone`	Corrected lowercase smartphone entries to capitalized format
3	Category	`Electronicscscs / Electroni`	`Electronics`	Cleaned 6 corrupted/truncated category spelling errors

2. COMPLETE DATASET (POST-TRANSFORMATION)

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
1	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	130	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics	900	Electronics
5	22-MAY-US	Backpack	North Face	70	20		70	Electronics
6	07-JUN-UK	Headphones	Sony		45	Electronics	309	Electronics
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40		90	Electronics
13	08-DEC-DE	Coffee Maker	Nespresso	120	35		120	Electronics
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	150	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	160	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronics	980	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	150	Fashion
20	14-AUG-US	Camping Tent	Coleman		10	Outdoor	309	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics	700	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	80	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30		150	Electronics
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics	800	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	309	Fashion

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	400	Kitchen
28	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
29	07-MAR-CA	Dress	Zara	60	30	Fashion	60	Fashion
30	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	40	Kitchen
31	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	130	Electronics
32	02-DEC-CA	Jeans	Levi's	50	50	Fashion	50	Fashion
33	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
34	09-JUL-FR	Watch	Casio	100	20	Accessories	100	Accessories

Assignment 2: Data Cleaning and Transformation

Dataset Report & Transformation Audit Log

TASK: CORRECTING INCONSISTENT DATA

- **Identify Inconsistent Text Formats:** Detect inconsistent text capitalization present in the *"Product Name"* column (e.g., lowercase `headphones`, `smartphone` vs. Proper Case).
- **Identify Typos in Category:** Locate misspellings and truncated text in the *"Category"* column (e.g., `Electronicscscs`, `Electroni`).
- **Standardize via Find & Replace:** Use Excel's Find and Replace functionality to standardize all product names to Proper Case and correct all category typos.

1. FIND & REPLACE OPERATIONS LOG

STEP	TARGET COLUMN	FIND WHAT	REPLACE WITH	PURPOSE / RESULT
1	Product Name	`headphones`	`Headphones`	Standardized lowercase product entries to Proper Title Case
2	Product Name	`smartphone`	`Smartphone`	Corrected lowercase smartphone entries to capitalized format
3	Category	`Electronicscscs` / `Electroni`	`Electronics`	Cleaned 6 corrupted/truncated category spelling errors across cells

2. COMPLETE DATASET (POST-TRANSFORMATION)

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
1	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	130	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics	900	Electronics
5	22-MAY-US	Backpack	North Face	70	20		70	Electronics
6	07-JUN-UK	Headphones	Sony		45	Electronics	309	Electronics
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40		90	Electronics
13	08-DEC-DE	Coffee Maker	Nespresso	120	35		120	Electronics
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	150	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	160	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronics	980	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	150	Fashion
20	14-AUG-US	Camping Tent	Coleman		10	Outdoor	309	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics	700	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	80	Kitchen

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30		150	Electronics
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics	800	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	309	Fashion
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	400	Kitchen
28	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
29	07-MAR-CA	Dress	Zara	60	30	Fashion	60	Fashion
30	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	40	Kitchen
31	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	130	Electronics
32	02-DEC-CA	Jeans	Levi's	50	50	Fashion	50	Fashion
33	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
34	09-JUL-FR	Watch	Casio	100	20	Accessories	100	Accessories

Power Query Editor: Duplicate Removal Report

Query Name: **table_products** — Sequential Deduplication Audit

REMOVING DUPLICATES:

- Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

Transformation Overview: Seven sequential `Table.Distinct` steps were applied in Power Query Editor to remove duplicate values across each column, reducing the dataset to 3 distinct records.

1. APPLIED STEPS & M FORMULAS

STEP	APPLIED STEP NAME	TARGET COLUMN	POWER QUERY M FORMULA	ROWS
1	Removed Duplicates	Product ID	`= Table.Distinct(#"Changed Type", {"Product ID"})`	29+
2	Removed Duplicates1	Product Name	`= Table.Distinct(#"Removed Duplicates", {"Product Name"})`	22
3	Removed Duplicates2	Brand Name	`= Table.Distinct(#"Removed Duplicates1", {"Brand Name"})`	22
4	Removed Duplicates3	Price ($)	`= Table.Distinct(#"Removed Duplicates2", {"Price ($)"})`	16
5	Removed Duplicates4	Quantity	`= Table.Distinct(#"Removed Duplicates3", {"Quantity"})`	10
6	Removed Duplicates5	Category	`= Table.Distinct(#"Removed Duplicates4", {"Category"})`	4
7	Removed Duplicates6	Category(IMPUTED)	`= Table.Distinct(#"Removed Duplicates5", {"Category(IMPUTED)"})`	3

2. INTERMEDIATE OUTPUT (REMOVED DUPLICATES4 — 10 ROWS)

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	
6	07-JUN-UK	Headphones	Sony	*null*	45	Electronics
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	13-APR-CA	Toaster	Hamilton	40	10	Kitchen

3. FINAL DEDUPLICATED OUTPUT (REMOVED DUPLICATES6 — 3 ROWS)

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen

Splitting and Merging Data Report

Excel Formula Transformation & Complete Dataset

SPLITTING AND MERGING DATA:

- Split the *"Product ID"* column into two separate columns for *"Manufacturing Date"* and *"Country Code"*. Remove unnecessary characters, if any.
- Merge the *"Brand Name"* and *"Product Name"* columns into one column named *"Product Brand"*.

1. EXCEL FORMULAS APPLIED

STEP	TARGET COLUMN	EXCEL FORMULA	OPERATION DESCRIPTION
1	Manufacturing date (Col B)	`=LEFT(A2,2)`	Extracted the 2-digit manufacturing day from the start of the Product ID, removing hyphens and month text.
2	Country code (Col C)	`=RIGHT(A2,2)`	Extracted the 2-letter country code from the end of the Product ID.
3	Product Brand (Col E)	`=CONCATENATE(D2," ",F2)`	Merged Product Name (Col D) and Brand Name (Col F) with a space separator.

2. COMPLETE TRANSFORMED DATASET (34 ROWS)

#	PRODUCT	MFG DATE	COUNTRY	PRODUCT NAME	PRODUCT BRAND	BRAND NAME	PRICE ($)	QTY	CATEGORY	PRICE (IMP.)	CATEGORY (IMP.)
1	28-JAN-US	28	US	Laptop	Laptop Dell	Dell	1000	30	Electronics	1000	Electronics
2	15-FEB-US	15	US	Sneakers	Sneakers Nike	Nike	80	15	Fashion	80	Fashion
3	03-MAR-US	03	US	Coffee Maker	Coffee Maker Keurig	Keurig	130	40	Kitchen	130	Kitchen
4	11-APR-US	11	US	Smartphone	Smartphone Samsung	Samsung	900	25	Electronics	900	Electronics
5	22-MAY-US	22	US	Backpack	Backpack North Face	North Face	70	20		70	Electronics
6	07-JUN-UK	07	UK	Headphones	Headphones Sony	Sony		45	Electronics	285	Electronics
7	19-JUL-UK	19	UK	T-shirt	T-shirt Adidas	Adidas	30	5	Fashion	30	Fashion
8	23-AUG-UK	23	UK	Blender	Blender Ninja	Ninja	90	35	Kitchen	90	Kitchen
9	05-SEP-UK	05	UK	Tablet	Tablet Apple	Apple	500	50	Electronics	500	Electronics
10	14-OCT-UK	14	UK	Hiking Boots	Hiking Boots Timberland	Timberland	130	10	Outdoor	130	Outdoor
11	17-JUN-IN	17	IN	Laptop	Laptop HP	HP	950	25	Electronics	950	Electronics
12	25-NOV-AU	25	AU	Sneakers	Sneakers Adidas	Adidas	90	40		90	Electronics
13	08-DEC-DE	08	DE	Coffee Maker	Coffee Maker Nespresso	Nespresso	120	35		120	Electronics

#	PRODUCT	MFG DATE	COUNTRY	PRODUCT NAME	PRODUCT BRAND	BRAND NAME	PRICE ($)	QTY	CATEGORY	PRICE (IMP.)	CATEGORY (IMP.)
14	18-FEB-CA	18	CA	Smartwatch	Smartwatch Fitbit	Fitbit	150	15	Electronics	150	Electronics
15	16-APR-ES	16	ES	headphones	headphones Bose	Bose	250	20	Electronics	250	Electronics
16	21-AUG-CA	21	CA	Laptop Bag	Laptop Bag Samsonite	Samsonite	50	35	Accessories	50	Accessories
17	20-AUG-CN	20	CN	Smartwatch	Smartwatch Huawei	Huawei	160	15	Electronics	160	Electronics
18	27-JAN-IT	27	IT	Laptop	Laptop Asus	Asus	980	10	Electronics	980	Electronics
19	01-MAR-UK	01	UK	Sunglasses	Sunglasses Oakley	Oakley	150	15	Fashion	150	Fashion
20	14-AUG-US	14	US	Camping Tent	Camping Tent Coleman	Coleman		10	Outdoor	289	Outdoor
21	14-MAY-RU	14	RU	Camera	Camera Nikon	Nikon	700	50	Electronics	700	Electronics
22	09-JAN-CA	09	CA	Microwave	Microwave Panasonic	Panasonic	80	20	Kitchen	80	Kitchen
23	19-JUL-BR	19	BR	Fitness Tracker	Fitness Tracker Xiaomi	Xiaomi	150	30		150	Electronics
24	21-AUG-CA	21	CA	Laptop Bag	Laptop Bag Samsonite	Samsonite	50	35	Accessories	50	Accessories
25	29-SEP-CA	29	CA	Smartphone	Smartphone Google	Google	800	45	Electronics	800	Electronics
26	03-JUN-CA	03	CA	Sunglasses	Sunglasses Ray-Ban	Ray-Ban		25	Fashion	248	Fashion
27	11-JUL-CA	11	CA	Blender	Blender Vitamix	Vitamix	400	40	Kitchen	400	Kitchen
28	16-APR-ES	16	ES	headphones	headphones Bose	Bose	250	20	Electronics	250	Electronics
29	07-MAR-CA	07	CA	Dress	Dress Zara	Zara	60	30	Fashion	60	Fashion
30	13-APR-CA	13	CA	Toaster	Toaster Hamilton	Hamilton	40	10	Kitchen	40	Kitchen
31	24-MAY-CA	24	CA	Fitness Tracker	Fitness Tracker Garmin	Garmin	130	5	Electronics	130	Electronics
32	02-DEC-CA	02	CA	Jeans	Jeans Levi's	Levi's	50	50	Fashion	50	Fashion
33	17-JUN-IN	17	IN	Laptop	Laptop HP	HP	950	25	Electronics	950	Electronics
34	09-JUL-FR	09	FR	Watch	Watch Casio	Casio	100	20	Accessories	100	Accessories

Conditional Formatting Report

Book1 — Sheet1: Price Color Scales & Category Highlighting

CONDITIONAL FORMATTING TASKS:

- Apply data bar or color scales conditional formatting in the *"Price"* column.
- Create a custom rule for conditional formatting in the *"Category"* column to highlight cells where the category is *"Electronics."*
- **Deliverables:** Submit a project folder containing the formatted Excel workbook and documentation.

1. CONDITIONAL FORMATTING RULES APPLIED

RULE	TARGET COLUMN	FORMATTING TYPE	RULE / CONDITION DETAILS
1	Price(IMPUTED) (Col J)	Color Scale & Icon Set	3-Color Scale (Green for low prices $30–$80, Peach/Orange for mid-range $90–$500, Red for high prices $700–$1000).
2	Category(IMPUTED) (Col K)	Custom Highlight Rule	Text that Contains / Equal To **"Electronics"** highlighted with light red fill.

2. COMPLETE FORMATTED DATASET (34 ROWS)

#	MFG DATE	COUNTRY	PRODUCT NAME	PRODUCT BRAND	BRAND NAME	PRICE ($)	QTY	CATEGORY	PRICE (IMP.)	CATEGORY (IMP.)
1	28	US	Laptop	Laptop Dell	Dell	1000	30	Electronics	1000	Electronics
2	15	US	Sneakers	Sneakers Nike	Nike	80	15	Fashion	80	Fashion
3	03	US	Coffee Maker	Coffee Maker Keurig	Keurig	130	40	Kitchen	130	Kitchen
4	11	US	Smartphone	Smartphone Samsung	Samsung	900	25	Electronics	900	Electronics
5	22	US	Backpack	Backpack North Face	North Face	70	20		70	Electronics
6	07	UK	Headphones	Headphones Sony	Sony		45	Electronics	285	Electronics
7	19	UK	T-shirt	T-shirt Adidas	Adidas	30	5	Fashion	30	Fashion
8	23	UK	Blender	Blender Ninja	Ninja	90	35	Kitchen	90	Kitchen
9	05	UK	Tablet	Tablet Apple	Apple	500	50	Electronics	500	Electronics
10	14	UK	Hiking Boots	Hiking Boots Timberland	Timberland	130	10	Outdoor	130	Outdoor
11	17	IN	Laptop	Laptop HP	HP	950	25	Electronics	950	Electronics
12	25	AU	Sneakers	Sneakers Adidas	Adidas	90	40		90	Electronics
13	08	DE	Coffee Maker	Coffee Maker Nespresso	Nespresso	120	35		120	Electronics

#	MFG DATE	COUNTRY	PRODUCT NAME	PRODUCT BRAND	BRAND NAME	PRICE ($)	QTY	CATEGORY	PRICE (IMP.)	CATEGORY (IMP.)
14	18	CA	Smartwatch	Smartwatch Fitbit	Fitbit	150	15	Electronics	150	Electronics
15	16	ES	headphones	headphones Bose	Bose	250	20	Electronics	250	Electronics
16	21	CA	Laptop Bag	Laptop Bag Samsonite	Samsonite	50	35	Accessories	50	Accessories
17	20	CN	Smartwatch	Smartwatch Huawei	Huawei	160	15	Electronics	160	Electronics
18	27	IT	Laptop	Laptop Asus	Asus	980	10	Electronics	980	Electronics
19	01	UK	Sunglasses	Sunglasses Oakley	Oakley	150	15	Fashion	150	Fashion
20	14	US	Camping Tent	Camping Tent Coleman	Coleman		10	Outdoor	289	Outdoor
21	14	RU	Camera	Camera Nikon	Nikon	700	50	Electronics	700	Electronics
22	09	CA	Microwave	Microwave Panasonic	Panasonic	80	20	Kitchen	80	Kitchen
23	19	BR	Fitness Tracker	Fitness Tracker Xiaomi	Xiaomi	150	30		150	Electronics
24	21	CA	Laptop Bag	Laptop Bag Samsonite	Samsonite	50	35	Accessories	50	Accessories
25	29	CA	Smartphone	Smartphone Google	Google	800	45	Electronics	800	Electronics
26	03	CA	Sunglasses	Sunglasses Ray-Ban	Ray-Ban		25	Fashion	248	Fashion
27	11	CA	Blender	Blender Vitamix	Vitamix	400	40	Kitchen	400	Kitchen
28	16	ES	headphones	headphones Bose	Bose	250	20	Electronics	250	Electronics
29	07	CA	Dress	Dress Zara	Zara	60	30	Fashion	60	Fashion
30	13	CA	Toaster	Toaster Hamilton	Hamilton	40	10	Kitchen	40	Kitchen
31	24	CA	Fitness Tracker	Fitness Tracker Garmin	Garmin	130	5	Electronics	130	Electronics
32	02	CA	Jeans	Jeans Levi's	Levi's	50	50	Fashion	50	Fashion
33	17	IN	Laptop	Laptop HP	HP	950	25	Electronics	950	Electronics
34	09	FR	Watch	Watch Casio	Casio	100	20	Accessories	100	Accessories

Product Inventory & Imputed Data

Worksheet: Sheet1 (Rows 2–35)

Number Formatting:

- Format the data type of the “Price” column to a currency format.
- Format the “Manufacturing Date” column to display dates in the “DD-MM-YYYY ”

Manufacturing Date Formula (Column B): `=DATEVALUE(LEFT(A2,6)&"-2026")`

Product	Manufacturing date	Country code	Product Name	Product Brand	Brand Name	Price ($)	Quantity	Category	Price(IMPUTED)	Category(IMPUTED)
28-JAN-US	28-Jan-26	US	Laptop	Laptop Dell	Dell	1000	30	Electronics	$1,000.00	Electronics
15-FEB-US	15-Feb-26	US	Sneakers	Sneakers Nike	Nike	80	15	Fashion	$80.00	Fashion
03-MAR-US	3-Mar-26	US	Coffee Maker	Coffee Maker Keurig	Keurig	130	40	Kitchen	$130.00	Kitchen
11-APR-US	11-Apr-26	US	Smartphone	Smartphone Samsung	Samsung	900	25	Electronics	$900.00	Electronics
22-MAY-US	22-May-26	US	Backpack	Backpack North Face	North Face	70	20		$70.00	Electronics
07-JUN-UK	7-Jun-26	UK	Headphones	Headphones Sony	Sony		45	Electronics	$285.00	Electronics
19-JUL-UK	19-Jul-26	UK	T-shirt	T-shirt Adidas	Adidas	30	5	Fashion	$30.00	Fashion
23-AUG-UK	23-Aug-26	UK	Blender	Blender Ninja	Ninja	90	35	Kitchen	$90.00	Kitchen

Product	Manufacturing date	Country code	Product Name	Product Brand	Brand Name	Price ($)	Quantity	Category	Price(IMPUTED)	Category(IMPUTED)
05-SEP-UK	5-Sep-26	UK	Tablet	Tablet Apple	Apple	500	50	Electronics	$500.00	Electronics
14-OCT-UK	14-Oct-26	UK	Hiking Boots	Hiking Boots Timberland	Timberland	130	10	Outdoor	$130.00	Outdoor
17-JUN-IN	17-Jun-26	IN	Laptop	Laptop HP	HP	950	25	Electronics	$950.00	Electronics
25-NOV-AU	25-Nov-26	AU	Sneakers	Sneakers Adidas	Adidas	90	40		$90.00	Electronics
08-DEC-DE	8-Dec-26	DE	Coffee Maker	Coffee Maker Nespresso	Nespresso	120	35		$120.00	Electronics
18-FEB-CA	18-Feb-26	CA	Smartwatch	Smartwatch Fitbit	Fitbit	150	15	Electronics	$150.00	Electronics
16-APR-ES	16-Apr-26	ES	headphones	headphones Bose	Bose	250	20	Electronics	$250.00	Electronics
21-AUG-CA	21-Aug-26	CA	Laptop Bag	Laptop Bag Samsonite	Samsonite	50	35	Accessories	$50.00	Accessories
20-AUG-CN	20-Aug-26	CN	Smartwatch	Smartwatch Huawei	Huawei	160	15	Electronics	$160.00	Electronics
27-JAN-IT	27-Jan-26	IT	Laptop	Laptop Asus	Asus	980	10	Electronics	$980.00	Electronics
01-MAR-UK	1-Mar-26	UK	Sunglasses	Sunglasses Oakley	Oakley	150	15	Fashion	$150.00	Fashion
14-AUG-US	14-Aug-26	US	Camping Tent	Camping Tent Coleman	Coleman		10	Outdoor	$289.00	Outdoor

Product	Manufacturing date	Country code	Product Name	Product Brand	Brand Name	Price ($)	Quantity	Category	Price(IMPUTED)	Category(IMPUTED)
14-MAY-RU	14-May-26	RU	Camera	Camera Nikon	Nikon	700	50	Electronics	$700.00	Electronics
09-JAN-CA	9-Jan-26	CA	Microwave	Microwave Panasonic	Panasonic	80	20	Kitchen	$80.00	Kitchen
19-JUL-BR	19-Jul-26	BR	Fitness Tracker	Fitness Tracker Xiaomi	Xiaomi	150	30		$150.00	Electronics
21-AUG-CA	21-Aug-26	CA	Laptop Bag	Laptop Bag Samsonite	Samsonite	50	35	Accessories	$50.00	Accessories
29-SEP-CA	29-Sep-26	CA	Smartphone	Smartphone Google	Google	800	45	Electronics	$800.00	Electronics
03-JUN-CA	3-Jun-26	CA	Sunglasses	Sunglasses Ray-Ban	Ray-Ban		25	Fashion	$248.00	Fashion
11-JUL-CA	11-Jul-26	CA	Blender	Blender Vitamix	Vitamix	400	40	Kitchen	$400.00	Kitchen
16-APR-ES	16-Apr-26	ES	headphones	headphones Bose	Bose	250	20	Electronics	$250.00	Electronics
07-MAR-CA	7-Mar-26	CA	Dress	Dress Zara	Zara	60	30	Fashion	$60.00	Fashion
13-APR-CA	13-Apr-26	CA	Toaster	Toaster Hamilton	Hamilton	40	10	Kitchen	$40.00	Kitchen
24-MAY-CA	24-May-26	CA	Fitness Tracker	Fitness Tracker Garmin	Garmin	130	5	Electronics	$130.00	Electronics
02-DEC-CA	2-Dec-26	CA	Jeans	Jeans Levi's	Levi's	50	50	Fashion	$50.00	Fashion
17-JUN-IN	17-Jun-26	IN	Laptop	Laptop HP	HP	950	25	Electronics	$950.00	Electronics
09-JUL-FR	9-Jul-26	FR	Watch	Watch Casio	Casio	100	20	Accessories	$100.00	Accessories